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CIVL4340
Wind Engineering

Week 10: Internal pressures & worked example

Reading

- Holmes, Chapters 5

Internal wind loading

Pressure loading (no internal drag loads).

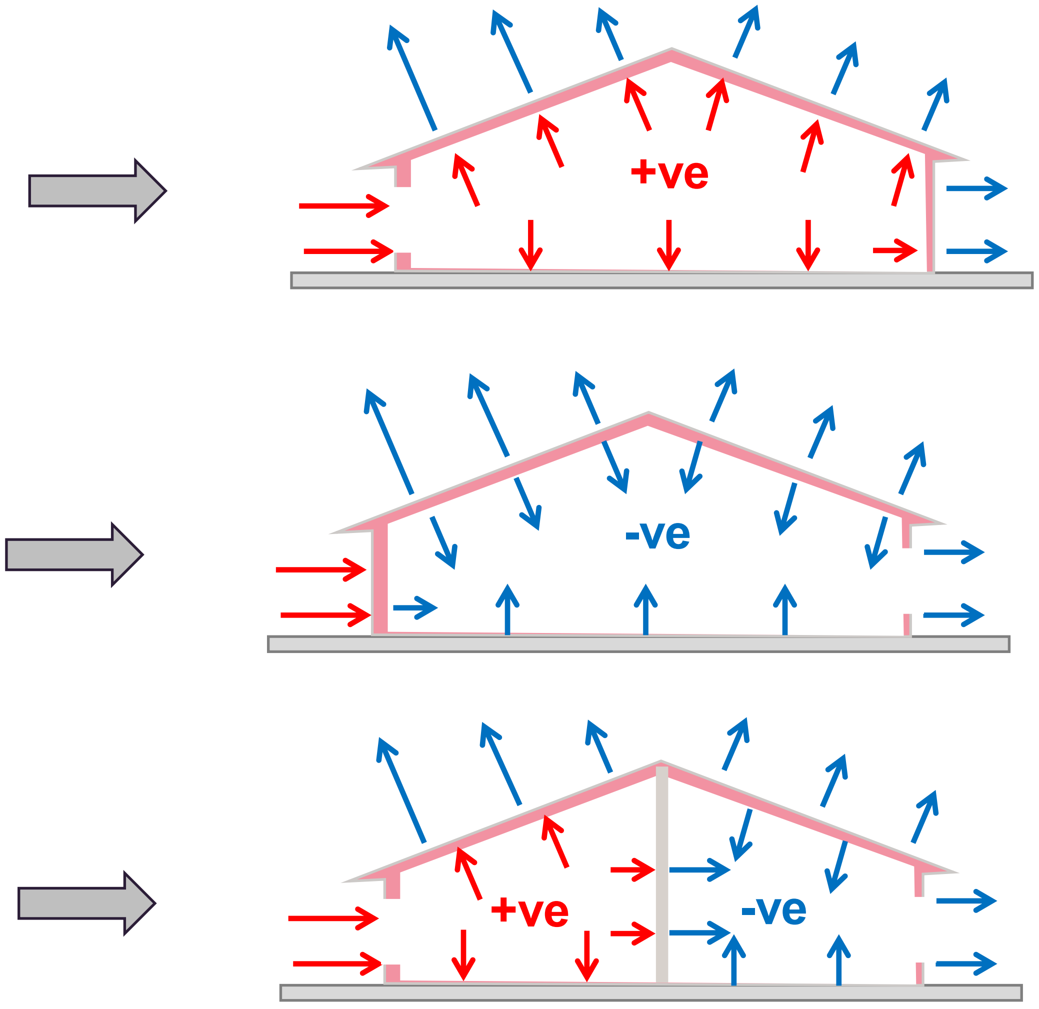
Can significantly increase structural loads – can be the primary load case in some instances.

Why do we get internal pressures?

A: Changes in air density within a building (can be maintained by very small mass flow rates)

- Approximately uniform on all internal surfaces.
- Approximately equal to external pressure, if a dominant opening exists, or an average of external pressure at all openings.

Internal pressures



Internal pressure failures (external)

Internal pressure failures (internal)

FEMA

www.wbdg.org

Internal pressures

Dimensional analysis allows us to determine that for a rigid building with a single opening,

$$C_{pi} = \frac{p_i - p_o}{\frac{1}{2}\rho_a \overline{U}^2} = F(\Phi_1, \Phi_2, \Phi_3, \Phi_4, \Phi_5)$$

Opening area

Internal volume

$$\Phi_1 = A^{3/2}/V_o$$

Ratio of specific heat of air

Speed of sound

$$\Phi_2 = a_s/\overline{U} = \sqrt{\gamma p_o/\rho_a}\,/\,\overline{U}$$

$$\Phi_3 = \rho_a \overline{U} A^{1/2}/\mu$$ (Re)

$$\Phi_4 = \sigma_u/\overline{U}$$ (Turbulence Intensity)

Integral length scale

$$\Phi_5 = l_u/\sqrt{A}$$

Internal pressures

In practice, some buildings are flexible and internal pressures are also dependent on the small changes in internal volume that this flexibility allows.

Response of internal pressure is not instantaneous but does occur over short time periods.

Time to reach equilibrium

Initial internal Cp

$$\tau = \frac{\rho_a V_o \bar{U}}{\gamma k A p_o} \sqrt{C_{pe} - C_{pio}}$$

Loss coefficient

Ambient pressure

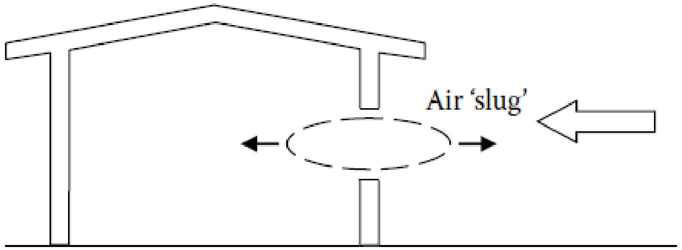
Example -> Holmes p151

Internal pressure

Internal pressures are dynamic (i.e. they fluctuate).

Due to:

- Turbulence in the wind
- Relative size of openings
- Helmholtz resonance (governs peak internal pressures)



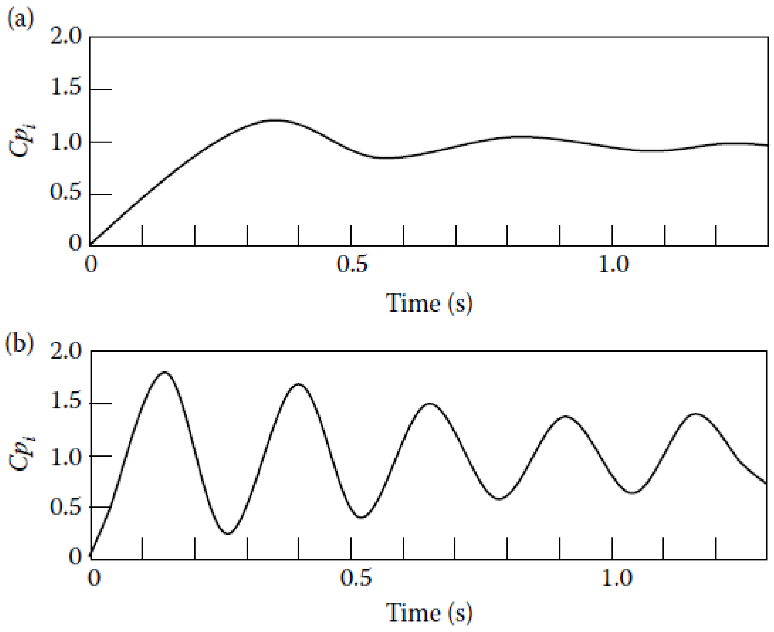


Figure 6.2 Response to a step change in external pressure, $V_o = 600$ m^3, $\overline{U} = 30$ m/s. (a) $A = 1$ m^2; (b) $A = 9$ m^2.

Internal pressure

Relating external to internal pressures:

$$\frac{\sigma_{pi}}{\sigma_{pe}} = 1.1 \quad \text{for } S^* \geq 1.0$$

$$\frac{\sigma_{pi}}{\sigma_{pe}} = 1.1 + 0.2\log_{10}(S^*) \quad \text{for } S^* < 1.0$$

where

$$S^* = (a_s/\overline{U})^2\left(A^{3/2}/V_0\right)$$

which can be converted to a peak relationship for openings on the windward wall through,

$$\frac{\hat{p}_i}{\hat{p}_e} = \frac{1 + 2gI_u\left(\sigma_{pi}/\sigma_{pe}\right)}{1 + 2gI_u}$$

Example -> Holmes p156-157

Internal pressures

Multiple openings (Mean pressures)

$$\sum_{1}^{N} \rho_a Q_j = 0$$

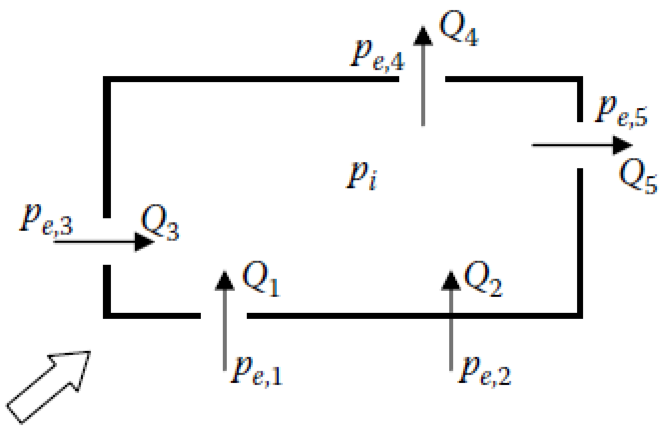
[Continuity]

which can be simplified to

$$\sum_{1}^{N} A_j \sqrt{|p_{e,j} - p_i|} = 0$$

So, as long as we know external pressures and the area of our openings we can calculate the internal pressure.

Internal pressure



$$A_1\sqrt{|p_{e,1} - p_i|} + A_2\sqrt{|p_{e,2} - p_i|} + A_3\sqrt{|p_{e,3} - p_i|} = A_4\sqrt{|p_{e,4} - p_i|} + A_5\sqrt{|p_{e,5} - p_i|}$$

Then solve for p_i.

You can also use Cp_i in place of p_i

C_{shp} – Internal [5.3]

- As with external pressure loads, internal pressures load requirements are prescribed using an internal aerodynamic shape factor, C_{shp}
- Only for enclosed buildings
- (Potential) openings must be able to resist design wind pressures to be considered sealed

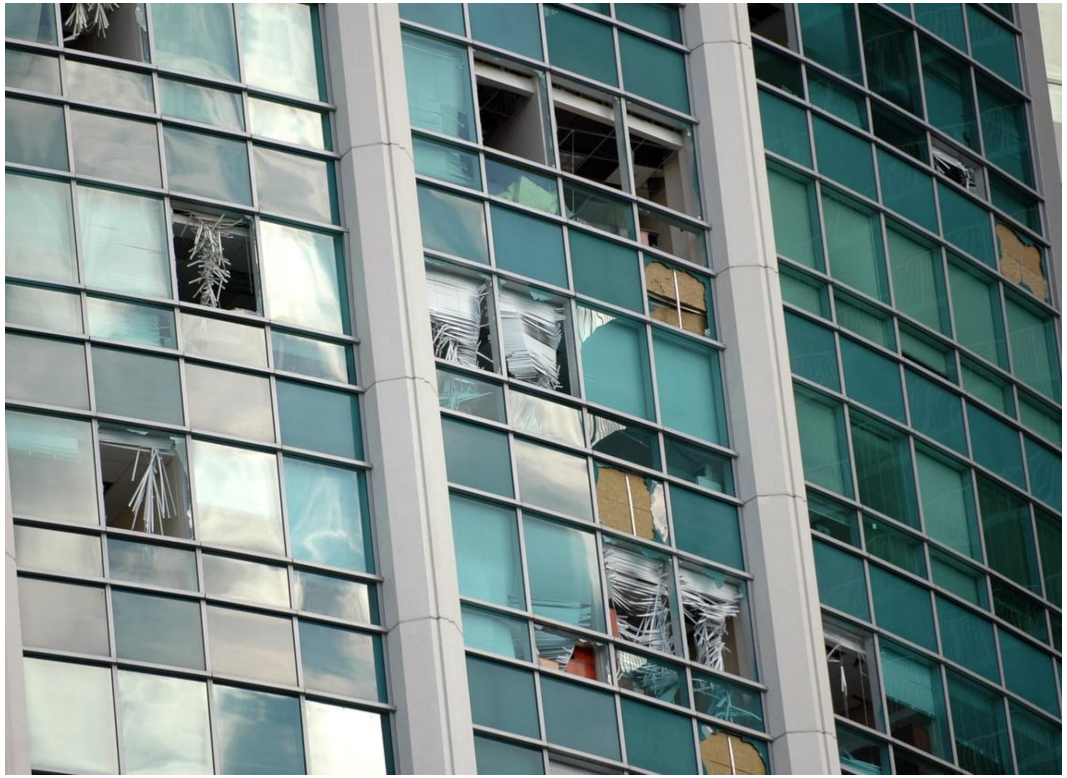
Internal pressure coefficient

Internal pressure combination factor

$$C_{shp} = C_{p,i} K_{c,i} K_v$$

Internal volume factor

Debris impact [2.5.8]



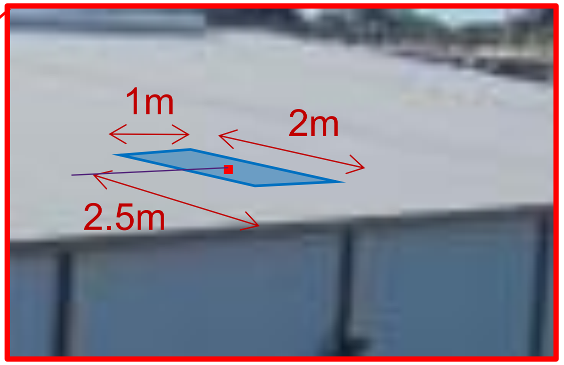
https://disastersafety.org/wp-content/uploads/2019/02/iStock-140268364.jpg

Texas Tech University Debris Impact Facility

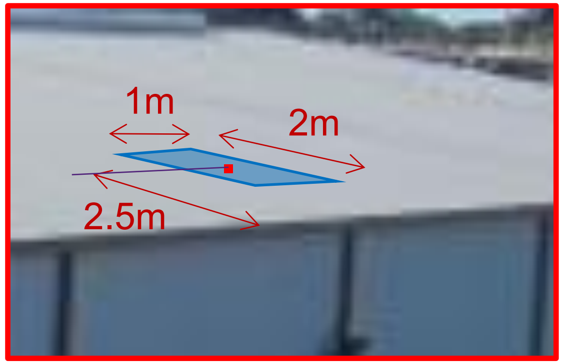
https://www.depts.ttu.edu/nwi/research/debrisimpact/index.php example

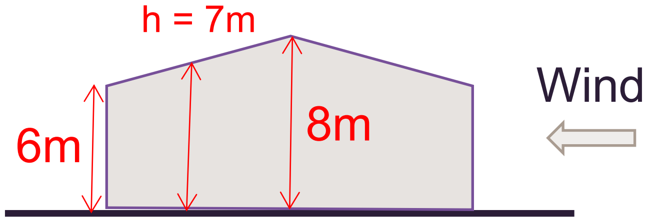
Example – external wind pressure

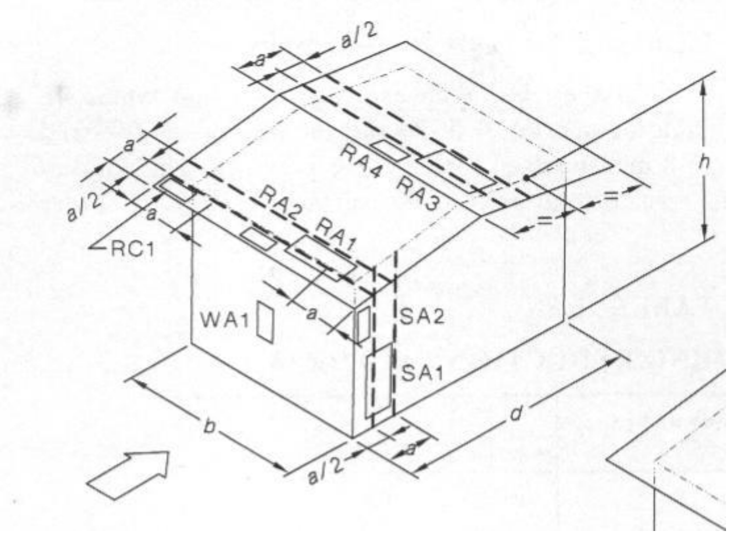
Determine the (external) wind pressure load a fixing holding the roof sheeting onto the industrial shed below must be designed for. The design wind speed is 59 m/s.



Worst case for wind







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$$p = (0.5\rho_{air})[V_{des,\theta}]^2 C_{shp} \cancelto{1}{C_{dyn}}$$

$$V_{des,\theta} = 59 m/s$$

$$C_{shp} = C_{shp,e} = C_{p,e} K_a K_{c,e} K_l \cancelto{1}{K_p}$$

$C_{p,e}$: [Table 5.3]

h/d = 7 m / 40 m = 0.175

$C_{p,e}$= -0.7, -0.3

K_a: [Table 5.4]

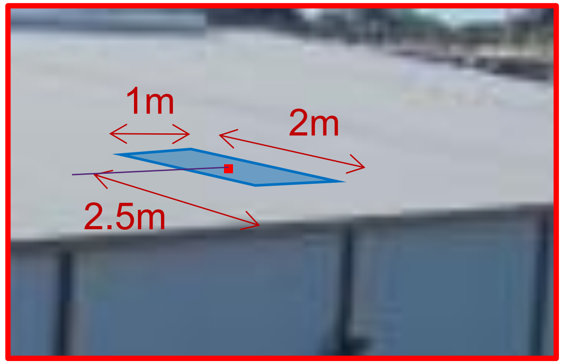
A = 2 m x 1 m = 2m^2 < 10 m^2

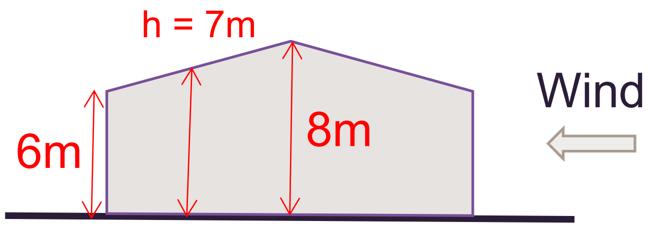
K_a = 1.0

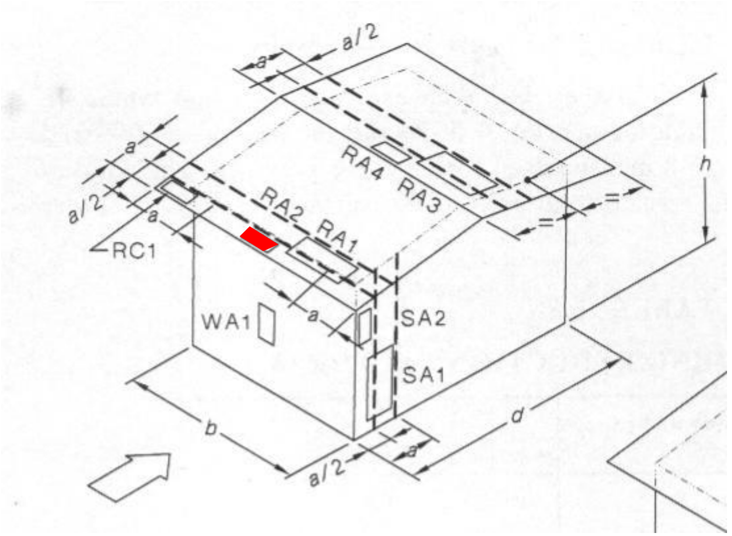
K_c = 1.0, but may be 0.9 if $C_{p,i}$ considered

K_l: [Table 5.6]

a = 2h *[both h/b,h/d<0.2]*
= 14 m







$a/2 = 7$ m
$a^2 = 196$ m^2
$0.25a^2 = 49$ m^2

$A = 2\text{m} \times 1\text{m} = 2\text{m}^2$
$A < 0.25a^2$
Trib area < a/2 from edge of roof

Therefore, RA2
$K_l = 2.0$

$$C_{shp} = C_{shp,e} = C_{p,e} K_a K_{c,e} K_l \cancelto{1}{K_p}$$

$$C_{shp} = -0.7 \times 1 \times 1 \times 2 \times 1 = -1.4$$

$$p = (0.5\,\rho_{air})\left[V_{des,\theta}\right]^2 C_{fig}\, C_{dyn}$$

$p = 0.6 \times 59^2 \times -1.4 \times 1$

$p = -2.9$ kPa

$\boxed{F = -2.9 \text{ kPa} \times 2 \text{ m}^2 = -5.8\text{kN}}$

Example – Internal wind pressure

Determine the internal wind pressure in the building below assuming it is in Region B1, wind is from the direction shown and the potential openings have areas as indicated. Ignore permeability and leakage on all faces. The design wind speed is 59 m/s.

Roller door: 20m^2
Door: 1.5m^2
Windows: 1m^2

Wind

Worst case (for roof loads)

All windows and doors on windward face fail, but none on the other faces (should test others)

$$p = (0.5\rho_{air})[V_{des,\theta}]^2 C_{shp} \cancelto{1}{C_{dyn}}$$

$$V_{des,\theta} = 59 m/s$$

$$C_{\text{shp}} = C_{\text{shp,i}} = C_{\text{p,i}} K_{\text{c,i}} K_{\text{v}}$$

Using Table 5.1(B)

Table 5.1(B) — Internal pressure coefficients ($C_{p,i}$) for buildings with openings greater than 0.5 % of the area of the corresponding wall or roof

Ratio of area of openings on one surface to the sum of the total open area (including permeability) of other wall and roof surfaces	Largest opening on windward wall	Largest opening on leeward wall	Largest opening on side wall	Largest opening on roof
0.5 or less	−0.3, 0.0	−0.3, 0.0	−0.3, 0.0	−0.3, 0.0
1	−0.1, 0.2	−0.3, 0.0	−0.3, 0.0	−0.3, 0.0
2	0.7 $K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$
3	0.85 $K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$
6 or more	$K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$	$K_a K_\ell C_{p,e}$
	t5-1(b)-1			

NOTE 1 $C_{p,e}$ is the relevant external pressure coefficient at the location of the largest opening. For example, in Column 2, $C_{p,e}$ means the windward wall pressure coefficient obtained from Table 5.2(A); in Column 3, $C_{p,e}$ means the leeward wall pressure coefficient obtained from Table 5.2(B), in Column 5, $C_{p,e}$ means the roof pressure coefficient for that part of the roof containing the opening.

NOTE 2 K_a is the area reduction factor related to the total area of the opening(s), A, on the surface under consideration treating the "tributary area" as the area of the opening. See Clause 5.4.2.

NOTE 3 K_ℓ is the local pressure factor, based on the area and location of the opening on the surface under consideration, treating the "Area, A" as the area of the opening. See Clause 5.4.4.

NOTE 4 Surfaces with openings have a ratio of total open area, A, to the total area of that surface related to the internal volume (Vol) under consideration, greater than 0.5 %.

Ratio of open area on windward face is > 6 times others

$$C_{p,i} = C_{p,e} K_l K_a$$

$$C_{p,e} = 0.7 \text{ [Table 5.2(A)]}$$

$$Area = (2*1.5) + (1*20) + (5*1) = 28\text{m}^2$$

$$K_a = 0.95 \text{ (rounded) [Table 5.4]}$$

$$a = 14 \text{ m (see previous example)}$$

$$K_l = 1.5 \text{ as Area} < 0.25a^2 \text{ [Table 5.6]}$$

$$C_{p,i} = 0.7*1.5*0.95 = 1$$

$$C_{shp} = C_{shp,i} = C_{p,i} K_{c,i} K_v$$

Let $K_{c,i}$ = 1 as we're just looking at internal pressure

Building volume = 50m * 40m * 7m = 14,000 m^3

Calculate K_v

100($A^{3/2}$/Vol) = 100*($28^{3/2}$/14,000) = 1.06

K_v = 1.01 + 0.15(log(100($A^{3/2}$/Vol))) = 1.01

$$C_{shp} = C_{p,i} K_{c,i} K_v = 1*1*1.01 = 1.01$$

$$p = (0.5\rho_{air})[V_{des,\theta}]^2 C_{shp} \cancelto{1}{C_{dyn}}$$

p = 0.6 x 59^2 x 1.01 x 1

$\boxed{\text{p = 2.1 kPa}}$

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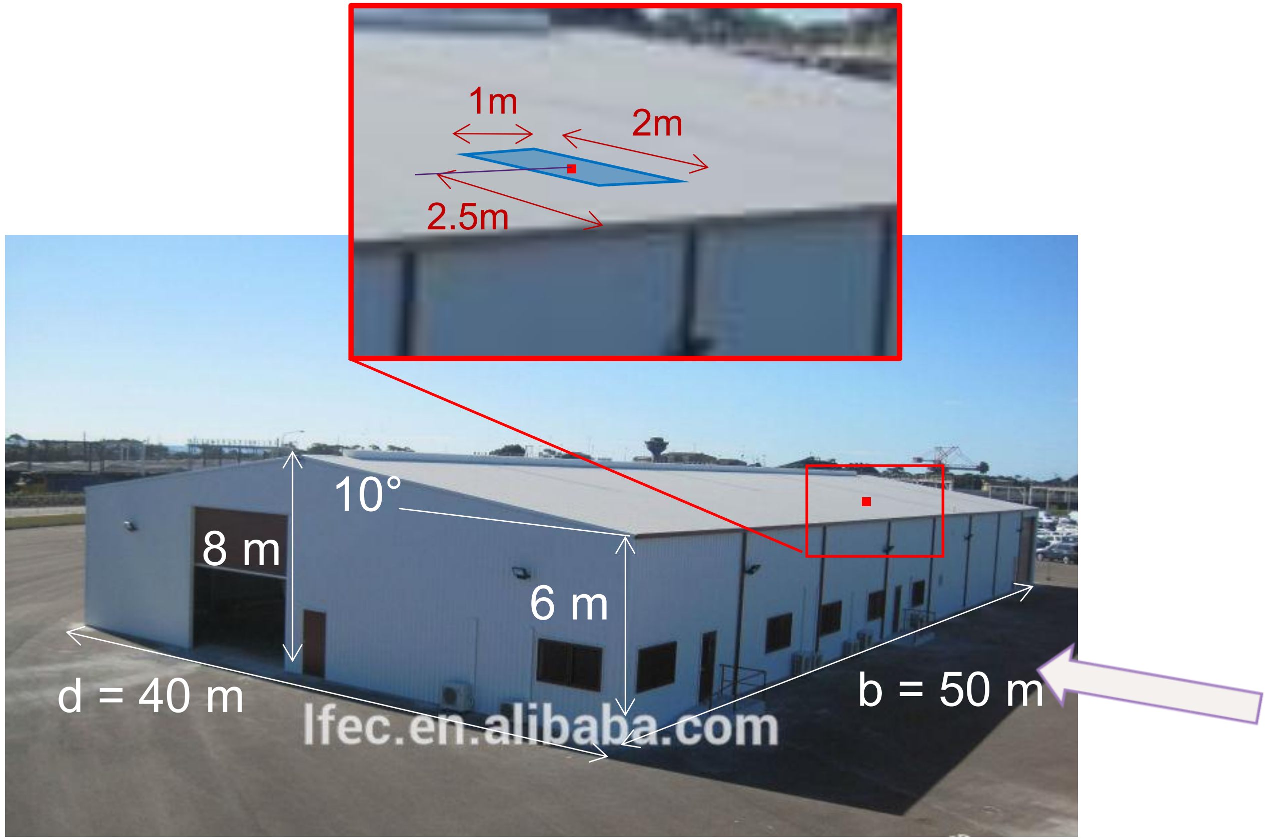
Example – Net pressure

To work out the net pressure on the cladding in the external pressure example we need to combine the internal and external pressures being careful to watch the sign convention and to include the combination factors, Kc.



Kc = 0.9 (two effective surfaces)

$F_{external}$ = -5.8 kN * 0.9 = -5.22 kN

$F_{internal}$ = 2.1kPa * 2m^2 * 0.9 = 3.8 kN

$\boxed{F_{net} = -5.8 \text{ kN} - 3.8 \text{ kN} = -9.6 \text{ kN (acting up)}}$

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